

Institute of Actuaries of India

Subject CT6 – Statistical Methods

May 2012 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1 :

- (i) The maximum loss for decisions d_1 , d_2 and d_3 are 18, 20 and 16, respectively. The minimax decision is d_3 .

- (ii) $P(\theta_1) = 0.25$, $P(\theta_2) = 0.45$, $P(\theta_3) = 0.30$.

Therefore,

Expected loss for d_1 is $0 \times 0.25 + 6 \times 0.45 + 18 \times 0.30 = 8.1$.

Expected loss for d_2 is $12 \times 0.25 + 0 \times 0.45 + 20 \times 0.30 = 9.0$.

Expected loss for d_3 is $16 \times 0.25 + 10 \times 0.45 + 0 \times 0.30 = 8.5$.

Therefore, the Bayes decision is d_1 .

[5]

Solution 2 :

- (i) By expressing the run-off triangle in terms of development years, we have

Accident Year	Development Year			
	0	1	2	3
2007	200	248	287	A
2008	215	C	290	
2009	234	260		

Development factor from year 2 to ultimate = $380/290 = 1.310345$.

Hence $A = 287 \times 1.310345 = 376.069$.

As all the claims are paid by the end of 3rd development year, $B = A = 376.07$.

Development factor from year 1 to ultimate = $405/260 = 1.557692$.

Development factor from year 1 to year 2 = $1.557692/1.310345 = 1.188765$.

Solving, $1.188765 = (287 + 290) / (248 + C)$,

i.e., $C = 237.38$.

- (ii) Development Factor for year 1 = $(248 + 237 + 260) / (200 + 215 + 234) = 1.148502$.

Development Factor to ultimate = $1.148502 \times 1.188765 \times 1.310345 = 1.789012$.

Thus, claims paid in year 2010 is

$450 - 400 \times 0.95 \times (1 - 1/1.789012) = 282.41$.

- (iii) Possible reasons include:

- Claim pattern for the year **might have** been affected by a catastrophe.
- Characteristics of the insured in the portfolio might have drastically changed.
- Rate of premium might have decreased in that year.
- Inflation for that year might have been unusually steep.

- Claim expenses (if included in calculating loss ratio) might have increased.

[10]

Solution 3 :

(i) The purposes of a practical time series analysis may be summarized as:

- Description of the data;
- Construction of a model which fits the data;
- Forecasting future values of the process;
- Deciding whether the process is out of control, requiring action;
- For vector time series, investigating connections between two or more observed processes with the aim of using values of some of the processes to predict those of the others.

(ii) AR(1) and random walk are Markov.

MA(1) and AR(2) are not Markov.

(iii) a. $E[X(t)] = a + bt + E[Y(t)]$.

Since $Y(t)$ is stationary, $E[Y(t)]$ is equal to some constant c that does not depend on t .

Therefore, $E[X(t)] = a + bt + c$, which depends on t .

It follows that $X(t)$ cannot be stationary.

b. $E[\nabla X(t)] = E[X(t)] - E[X(t-1)] = a + bt + c - [a + b(t-1) + c] = b$.

Thus, the mean does not depend on t .

$$\begin{aligned} \text{Cov}[\nabla X(t), \nabla X(s)] &= \text{Cov}[\{X(t) - X(t-1)\}, \{X(s) - X(s-1)\}] \\ &= \text{Cov}[\{b + Y(t) - Y(t-1)\}, \{b + Y(s) - Y(s-1)\}] \\ &= \text{Cov}[\{Y(t) - Y(t-1)\}, \{Y(s) - Y(s-1)\}] \\ &= \text{Cov}[Y(t), Y(s)] - \text{Cov}[Y(t), Y(s-1)] - \text{Cov}[Y(t-1), Y(s)] \\ &\quad + \text{Cov}[Y(t-1), Y(s-1)] \\ &= C_Y(t-s) - C_Y(t-s+1) - C_Y(t-s-1) + C_Y(t-s), \end{aligned}$$

where C_Y is the covariance function of the stationary process $Y(t)$. The last expression depends on t and s only through the difference $t-s$. Hence, the process $\nabla X(t)$ is stationary.

As $\nabla X(t)$ is stationary but $X(t)$ is not, the process $X(t)$ is $I(1)$.

(iv) Compute the difference between the processes:

$$\begin{aligned} X_1(t) - X_2(t) &= a[X_1(t-1) - X_2(t-1)] + b[X_2(t-1) - X_1(t-1)] \\ &\quad + [e_1(t) - e_2(t)] \\ &= (a-b)[X_1(t-1) - X_2(t-1)] + [e_1(t) - e_2(t)]. \end{aligned}$$

Let $Y(t) = X_1(t) - X_2(t)$, and $e(t) = e_1(t) - e_2(t)$.

Since $e_1(t)$ and $e_2(t)$ are white noise processes, $e(t)$ is also a white noise process.

Therefore, $Y(t)$ must be an AR(1) process.

$Y(t)$ is stationary as long as $|a - b| < 1$.

Since $X_1(t)$ and $X_2(t)$ are themselves $I(1)$, and their difference (i.e., a specific linear combination) is stationary whenever $|a - b| < 1$, the two processes are cointegrated whenever $|a - b| < 1$.

[11]

Solution 4 :

- (i) Let S_1 and S_2 denote the annual aggregate claims for the two portfolios. Then

$$\begin{aligned} E(S) &= E(S_1) + E(S_2) = 20 \times 10,000 + 30 \times 5,000 = 350,000. \\ Var(S) &= Var(S_1) + Var(S_2) = 20 \times 10,000^2 + 30 \times (5,000^2 + 5,000^2) \\ &= 3,500,000,000. \end{aligned}$$

- (ii) The required initial capital u is such that

$$P(u + c < S) = 0.05,$$

where c is the rate of continuously paid premium, i.e., $c = 1.15E(S)$.

This condition is equivalent to

$$P\left(\frac{S - E(S)}{\sqrt{Var(S)}} > \frac{u + c - E(S)}{\sqrt{Var(S)}}\right) = 0.05.$$

In view of the assumption of normal distribution, the above condition is equivalent to

$$\frac{u + c - E(S)}{\sqrt{Var(S)}} = 1.645,$$

$$\text{i.e., } u = 1.645 \times \sqrt{Var(S)} + E(S) - 1.15 E(S),$$

$$\text{i.e., } u = 1.645 \times \sqrt{Var(S)} - 0.15E(S) = 44,820.$$

Hence, the requisite initial capital is Rs. 44,820.

- (iii) Let S' be the modified annual aggregate claims under reinsurance, and u' be the requisite initial capital.

For proportional reinsurance with 80% retention,

$$S' = 0.8 \times S, \quad E(S') = 0.8E(S) \quad \text{and} \quad Var(S') = (0.8)^2 Var(S).$$

Let c' be the premium paid by the insurer to the reinsurer,

$$\text{i.e., } c' = (1 + \epsilon) \times 0.2 \times E(S).$$

The initial capitals u' and u (with and without reinsurance) should be such that the probability of ruin is equal to both $P(u' + c - c' < S')$ and $P(u + c < S)$. In view of the assumption of normal distribution, the normalized thresholds

$$\frac{u' + c - c' - E(S')}{\sqrt{Var(S')}} \quad \text{and} \quad \frac{u + c - E(S)}{\sqrt{Var(S)}},$$

must correspond to the same quintile of the standard normal distribution. Thus,

$$\frac{u' + c - c' - E(S')}{\sqrt{\text{Var}(S')}} = \frac{u + c - E(S)}{\sqrt{\text{Var}(S)}}.$$

It follows that

$$\frac{u' + 1.15E(S) - (1 + \varepsilon) \times 0.2 \times E(S) - 0.8 \times E(S)}{0.8\sqrt{\text{Var}(S)}} = \frac{u + 1.15E(S) - E(S)}{\sqrt{\text{Var}(S)}}.$$

Simplification of this equation leads to the following series of conditions.

$$u' = 0.8 \times (u + 0.15E(S)) - 1.15E(S) + (1 + \varepsilon) \times 0.2 \times E(S) + 0.8 \times E(S),$$

$$u' = 0.8u + (0.8 \times 0.15 - 1.15 + 0.2 + 0.2\varepsilon + 0.8)E(S),$$

$$u' = 0.8u + (0.2\varepsilon - 0.03)E(S).$$

[The equation given in the question has 0.03 printed as 0.83, which is incorrect. Credit should be given to candidates who have written equations correctly before simplification.]

- (iv) The value of u has already been obtained in part (ii). By equating the probability of ruin under reinsurance to 0.05, we have, under the assumption of normal distribution of S' ,

$$\frac{u' + 1.15E(S) - (1 + \varepsilon) \times 0.2 \times E(S) - 0.8 \times E(S)}{0.8\sqrt{\text{Var}(S)}} = 1.645,$$

$$\text{i.e., } u' = 1.645 \times 0.8\sqrt{\text{Var}(S)} - 1.15E(S) + (1 + \varepsilon) \times 0.2 \times E(S) + 0.8 \times E(S),$$

i.e., $u' = 25,356 + 70,000\varepsilon$. This would be smaller than 44,820, i.e., the answer obtained in part (ii), if and only if $\varepsilon < 0.2781$.

Alternatively, from part (iii), $u' < u$ if and only if $\varepsilon < \frac{1}{0.2} \left(\frac{0.2u}{E(S)} + 0.03 \right) = 0.2781$.

[12]

Solution 5 :

- The number of claims from a risk cannot be more than 1.
- The probability of claim from different risks can be different.
- Claim amounts from the risks need not be identically distributed.
- The number of risks is fixed (non-random).
- It is the risks, and not the multiple claims arising out of a particular risk, that have to be independent.

[3]

Solution 6 :

- (i) The likelihood function is

$$f(x_1, x_2, x_3, x_4, x_5 | \theta) = \prod_{i=1}^5 \binom{n_i}{x_i} \theta^{x_i} (1 - \theta)^{n_i - x_i} \propto \theta^{\sum_{i=1}^5 x_i} (1 - \theta)^{\sum_{i=1}^5 n_i - x_i},$$

where the n_i 's and x_i 's are as in the given data table.

- (ii) The prior density function $\pi(\theta)$ is proportional to $\theta^{\alpha-1}(1-\theta)^{\beta-1}$.

Therefore, the posterior density function $f(\theta|x_1, x_2, x_3, x_4, x_5)$ is proportional to

$$\theta^{\alpha-1}(1-\theta)^{\beta-1} \times \prod_{i=1}^5 \theta^{x_i}(1-\theta)^{n_i-x_i} = \theta^{\alpha-1+\sum_{i=1}^5 x_i}(1-\theta)^{\beta-1+\sum_{i=1}^5 (n_i-x_i)}.$$

Therefore, the posterior distribution is beta with parameters

$$\alpha + \sum_{i=1}^5 x_i = 504 \text{ and } \beta + \sum_{i=1}^5 (n_i - x_i) = 4643.$$

- (iii) Since posterior is of the same form as the prior, the prior can be said to be conjugate.
- (iv) The Bayes estimator with respect to the zero/one loss function is the mode of the beta distribution obtained in part (ii).

The derivative of the log-posterior is

$$\frac{\alpha - 1 + \sum_{i=1}^5 x_i}{\theta} - \frac{\beta - 1 + \sum_{i=1}^5 (n_i - x_i)}{1 - \theta},$$

which is a decreasing function of θ , and becomes 0 at the unique value

$$\frac{\alpha - 1 + \sum_{i=1}^5 x_i}{\alpha + \beta - 2 + \sum_{i=1}^5 n_i}.$$

Therefore, the above expression must be the requisite mode.

For the present data, the estimate is $\frac{10-1+494}{10+20-2+5117} = 0.09776$.

- (v) The estimator of part (iv) can be written as

$$\frac{\alpha - 1 + \sum_{i=1}^5 x_i}{\alpha + \beta - 2 + \sum_{i=1}^5 n_i} = Z \times \frac{\sum_{i=1}^5 x_i}{\sum_{i=1}^5 n_i} + (1 - Z) \times \frac{\alpha - 1}{\alpha + \beta - 2},$$

where $\frac{\alpha-1}{\alpha+\beta-2}$ is the corresponding estimate based only on the prior (collateral data), $(\sum_{i=1}^5 x_i)/(\sum_{i=1}^5 n_i)$ is the estimate based only on the relevant risk, and

$$Z = \frac{\sum_{i=1}^5 n_i}{\alpha + \beta - 2 + \sum_{i=1}^5 n_i} = \frac{5117}{5145}$$

is the credibility factor.

[7]

Solution 7 :

- (i) We have

$$\begin{aligned} 1 &= \int_0^\infty f(x)dx = \int_0^1 f(x)dx + \int_1^\infty f(x)dx = k \left(\int_0^1 x^{-\frac{1}{2}}dx + \int_1^\infty e^{-x}dx \right) \\ &= k \left(2x^{\frac{1}{2}} \Big|_0^1 - e^{-x} \Big|_1^\infty \right) = k(2 + e^{-1}). \end{aligned}$$

Therefore, $k = (2 + e^{-1})^{-1} = 0.4223$.

(ii) The distribution function is $F(x) = \int_0^x f(y)dy$.

For the range $0 \leq x < 1$, $F(x) = k \int_0^x y^{-\frac{1}{2}} dy = 2kx^{\frac{1}{2}}$.

For the range $x \geq 1$, $F(x) = 1 - k \int_x^\infty e^{-y} dy = 1 - ke^{-x}$.

It follows that

$$F^{-1}(u) = \begin{cases} \left(\frac{u}{2k}\right)^2 & \text{for } 0 \leq u < 2k, \\ -\ln\left(\frac{1-u}{k}\right) & \text{for } 2k \leq u < 1. \end{cases}$$

Steps for generating samples from $f(x)$:

Step I. Generate a pseudo-random number U from the uniform distribution over.

Step II. Define

$$X = \begin{cases} \left(\frac{U}{0.8446}\right)^2 & \text{for } 0 \leq U < 0.8446, \\ -\ln\left(\frac{1-U}{0.4223}\right) & \text{for } 0.8446 \leq U < 1. \end{cases}$$

The number X computed in this way is a pseudo-random sample from $f(x)$.

(iii) By using the above steps, the four given samples from the uniform distribution produce the following samples from $f(x)$: 0.846, 0.058, 1.492, 6.046.

[8]

Solution 8 :

Let the Household, Shopkeeper and Auto portfolios be designated by suffixes H , S and A , and years be indexed by j . We have

$$\bar{X}_H = 587, \bar{X}_S = 743, \bar{X}_A = 929;$$

$$S_H^2 = \frac{1}{4} \sum_{j=1}^5 (X_{Hj} - \bar{X}_H)^2 = 163^2,$$

$$S_S^2 = \frac{1}{4} \sum_{j=1}^5 (X_{Sj} - \bar{X}_S)^2 = 211^2,$$

$$S_A^2 = \frac{1}{4} \sum_{j=1}^5 (X_{Aj} - \bar{X}_A)^2 = 231^2.$$

The overall mean is

$$\bar{X} = \frac{1}{3} (\bar{X}_H + \bar{X}_S + \bar{X}_A) = \frac{587 + 743 + 929}{3} = 753.$$

Further,

$$E(s^2(\theta)) = \frac{1}{3} (S_H^2 + S_S^2 + S_A^2) = \frac{163^2 + 211^2 + 231^2}{3} = 41483.67.$$

$$\begin{aligned}
 \text{Var}(m(\theta)) &= \frac{1}{2}[(\bar{X}_H - \bar{X})^2 + (\bar{X}_S - \bar{X})^2 + (\bar{X}_A - \bar{X})^2] - \frac{1}{5}E(S^2(\theta)) \\
 &= \frac{1}{2}[(587 - 753)^2 + (743 - 753)^2 + (929 - 753)^2] - 41483.67/5 \\
 &= 29,316 - 8,296.73 = 21,019.27.
 \end{aligned}$$

Thus, the credibility factor is

$$Z = \frac{5}{5 + \frac{41483.67}{21019.27}} = 0.71699.$$

Finally, the credibility premia (in Rs.'000s) for the three risks under EBCT Model 1 are as under.

Household	:	$0.71699 \times 587 + (1 - 0.71699) \times 753 = 634.0;$
Shopkeeper	:	$0.71699 \times 743 + (1 - 0.71699) \times 753 = 745.8;$
Auto	:	$0.71699 \times 929 + (1 - 0.71699) \times 753 = 879.2.$

[6]

Solution 9 :

(i) The probability mass function of the Poisson distribution is

$$f(y) = \frac{e^{-\mu} \mu^y}{y!}, y = 0, 1, 2, \dots.$$

This has to be written in the form

$$g(y) = \exp \left[\left(\frac{y\theta - b(\theta)}{a(\varphi)} \right) + c(y, \varphi) \right].$$

By rearranging the expression for $f(y)$, we have

$$f(y) = \exp \left[\left(\frac{y \log \mu - \mu}{1} \right) - \log y! \right].$$

It is clear that this expression has the form of $g(y)$, with

$$\theta = \log \mu,$$

$$b(\theta) = \mu = \exp(\theta),$$

$$a(\varphi) = 1,$$

$$c(y, \varphi) = -\log y!.$$

(ii) $E(Y) = b'(\theta) = \exp(\theta) = \mu.$

$$\text{Var}(Y) = a(\varphi)b''(\theta) = \exp(\theta) = \mu.$$

(iii) The canonical link function is $\theta = \log \mu$. Therefore, the surmised relation is

$$\log \mu = \theta = a + bx,$$

$$\text{i.e., } \mu = \exp(a + bx).$$

(iv) The likelihood is

$$\prod_{i=1}^n \left(\frac{e^{-\mu} \mu^{N_i}}{N_i!} \right)_{\mu=\exp(a+bx_i)}.$$

Therefore, the log-likelihood for the parameters a and b is

$$\begin{aligned} l(a, b) &= \sum_{i=1}^n \log \left(\frac{e^{-\exp(a+bx_i)} e^{N_i(a+bx_i)}}{N_i!} \right) \\ &= - \sum_{i=1}^n [e^{a+bx_i} - N_i(a + bx_i)] - \sum_{i=1}^n \log(N_i!). \end{aligned}$$

By differentiating $l(a, b)$ with respect to a and b , respectively, we have

$$\begin{aligned} \frac{\partial l(a, b)}{\partial a} &= - \sum_{i=1}^n [e^{a+bx_i} - N_i], \\ \frac{\partial l(a, b)}{\partial b} &= - \sum_{i=1}^n [e^{a+bx_i} - N_i] x_i. \end{aligned}$$

By setting these partial derivatives equal to zero, we get the likelihood equations

$$\begin{aligned} \sum_{i=1}^n e^{a+bx_i} &= \sum_{i=1}^n N_i, \\ \sum_{i=1}^n x_i e^{a+bx_i} &= \sum_{i=1}^n x_i N_i. \end{aligned}$$

If the maximum likelihood estimators of a and b exist, they must satisfy these equations.

(v) a. The modified equation is

$$\mu = \exp(a + cz + bx),$$

where z is a binary variable representing the gender ($z = 0$ for male, $z = 1$ for female) and c is an additional parameter.

b. The modified equation is

$$\mu = \exp[a + cz + (b + dz)x],$$

where d is yet another parameter.

[Full credit may be given for the alternative forms

a. $\mu = \exp(a_z + bx)$ for part (a),

b. $\mu = \exp(a_z + b_z x)$ for part (b),

where a_0 and b_0 are parameters applicable for males, and a_1 and b_1 are parameters for females.]

[9]

Solution 10 :

- (i) Let n be the number of policies sold in a year and N be the random variable denoting the number of claims arising out of these n policies. Then, clearly N has the Poisson distribution with parameter $n/4$.

Let X denote the amount of individual claims and Y denotes the expenses associated with the claim.

S is the sum of the total annual aggregate claim amount and the total annual aggregate expenses. Being the sum of two compound Poisson random variables, S is itself a compound Poisson random variable.

Then the MGF of S is $M_S(t) = \exp \left[\frac{n}{4} (M_{X+Y}(t) - 1) \right] = \exp \left[\frac{n}{4} (M_X(t)M_Y(t) - 1) \right]$.

We have $M_X(t) = \exp \left(600t + \frac{1}{2} 50^2 t^2 \right) = \exp(600t + 1250t^2)$. Further,

$$M_Y(t) = E(e^{tY}) = \int_{40}^{80} \frac{1}{80 - 40} e^{ty} dy = \frac{1}{40t} (e^{80t} - e^{40t}).$$

Putting the three expressions together, we have

$$M_S(t) = \exp \left[\frac{n}{4} \left(\frac{1}{40t} e^{600t+1250t^2} (e^{80t} - e^{40t}) - 1 \right) \right].$$

- (ii) Recall that S is a compound Poisson random variable. Hence

$$E(S) = \frac{n}{4} \left(600 + \frac{40 + 80}{2} \right) = 165n.$$

Since X and Y are independent,

$$Var(S) = \frac{n}{4} E[(X + Y)^2] = \frac{n}{4} [E(X^2) + 2E(X)E(Y) + E(Y^2)].$$

We have $E(X) = 600$, $E(Y) = (40 + 80)/2 = 60$. Further,

$$E(X^2) = Var(X) + [E(X)]^2 = 50^2 + 600^2 = 362,500.$$

Likewise,

$$E(Y^2) = Var(Y) + [E(Y)]^2 = \frac{(80 - 40)^2}{12} + 60^2 = 3,733.33.$$

Also,

Putting these calculations together, we have

$$Var(S) = \frac{n}{4} [362,500 + 2 \times 600 \times 60 + 3,733.33] = 109,558.33n.$$

Total annual premium collected under the n policies = $180n$.

If S is assumed to be approximately normal, then

$$P(S \leq 180n) = P \left(\frac{S - 165n}{\sqrt{109558.33n}} \leq \frac{180n - 165n}{\sqrt{109558.33n}} \right) = P \left(Z \leq \frac{15n}{\sqrt{109558.33n}} \right),$$

where Z is Normal(0,1).

In order that this probability is at least 0.99, we must have

$$\frac{15n}{\sqrt{109558.33n}} \geq 2.3263,$$

This is equivalent to the requirement $n > 2635$. Thus, the insurer must sell at least 2636 policies to be 99% confident to make a profit from the portfolio.

[10]

Solution 11 :

- (i) The likelihood for the parameter c is

$$\prod_{i=1}^{1000} (2cX_i e^{-cX_i^2}).$$

The log-likelihood is

$$l(c) = 1000 \ln(2c) + \sum_{i=1}^{1000} \ln(X_i) - c \sum_{i=1}^{1000} X_i^2.$$

Differentiating $l(c)$ and setting the derivative equal to 0, we have the likelihood equation

$$\frac{1000}{c} - \sum_{i=1}^{1000} X_i^2 = 0,$$

which yields the unique solution

$$\hat{c} = \frac{1000}{\sum_{i=1}^{1000} X_i^2}.$$

Note that the second derivative of the likelihood function is $-1000/c^2$, which is *always* negative. This implies that the above solution to the likelihood equation is indeed the unique MLE.

- (ii) For the given data, the MLE of the parameter c has the value

$$\hat{c}_1 = \frac{1000}{\sum_{i=1}^{1000} X_i^2} = \frac{1000}{97,644,400,000} = \frac{1}{97,644,400} = 1.024 \times 10^{-8}.$$

- (iii) The mean of the distribution is

$$E(X) = \int_0^{\infty} xf(x)dx = \int_0^{\infty} x \times 2cx e^{-cx^2} dx.$$

Integrating by parts, we have

$$E(X) = -xe^{-cx^2} \Big|_0^{\infty} + \int_0^{\infty} e^{-cx^2} dx = 0 + \int_0^{\infty} e^{-\frac{x^2}{2\sigma^2}} dx = \sqrt{2\pi\sigma^2} \int_0^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} dx,$$

where $\sigma^2 = \frac{1}{2c}$. It follows that

$$E(X) = \sqrt{2\pi\sigma^2} \times \frac{1}{2} = \sqrt{\frac{\pi\sigma^2}{2}} = \sqrt{\frac{\pi}{4c}}.$$

[Full credit should be given for computing this correctly from the formula of mean of the Weibull distribution.]

Therefore, the method of moments estimator is obtained by solving the equation

$$\sqrt{\frac{\pi}{4c}} = 5120.5,$$

and the estimate is

$$\hat{c}_2 = \frac{\pi}{4 \times 5120.5^2} = 2.995 \times 10^{-8}.$$

(iv) The distribution function is

$$F(x) = \int_0^x f(y)dy = 1 - \int_x^\infty f(y)dy = 1 - \int_x^\infty 2cy e^{-cy^2} dy = 1 - e^{-cx^2}.$$

By setting this expression equal to half, we have an equation for the median m :

$$1 - e^{-cm^2} = \frac{1}{2}.$$

This equation yields the solution $m = \sqrt{\frac{\ln 2}{c}}$.

For the given data, the sample median is 5000. Therefore, the method of percentiles estimator of c satisfies the equation

$$\sqrt{\frac{\ln 2}{c}} = 5000,$$

and the solution is

$$\hat{c}_3 = \frac{\ln 2}{5000^2} = 2.772 \times 10^{-8}.$$

[9]

Solution 12 :

(i) Let X_0 be the random variable representing the total claim amount in the year just passed, and X_n be the random variable for the total annual claim amount in the n^{th} year from now.

X_0 follows the given pdf, and $X_n = (1.1)^n X_0$. Therefore,

$$\begin{aligned} P(X_n > x) &= P(X_0 > (1.1)^{-n}x) = \int_{(1.1)^{-n}x}^{\infty} f(y)dy = \int_{(1.1)^{-n}x}^{\infty} \frac{2(500)^2}{(y+500)^3} dy \\ &= \frac{(500)^2}{[(1.1)^{-n}x + 500]^2}, \end{aligned}$$

whenever $x > 0$.

The probability density function of X_n is the negative derivative of the above, i.e.,

$$\frac{2(500)^2(1.1)^{-n}}{[(1.1)^{-n}x + 500]^3} \text{ for } x > 0.$$

- (ii) The expected size of claim in the n^{th} year is

$$\begin{aligned} E(X_n) &= \int_0^{\infty} P(X_n > x) dx = \int_0^{\infty} \frac{(500)^2}{[(1.1)^{-n}x + 500]^2} dx = (1.1)^n \int_0^{\infty} \frac{(500)^2}{[y + 500]^2} dy \\ &= 500(1.1)^n. \end{aligned}$$

- (iii) The insurer's claim payment (Y_n) in the n^{th} year has the expected value

$$E(Y_n) = \int_0^{600} \frac{2(500)^2(1.1)^{-n}x}{[(1.1)^{-n}x + 500]^3} dx + 600P(X_n > 600).$$

The second term is

$$600P(X_n > 600) = \frac{600(500)^2}{[600(1.1)^{-n} + 500]^2} = \frac{600}{[1 + 1.2(1.1)^{-n}]^2}.$$

The first term simplifies as follows.

$$\begin{aligned} \int_0^{600} \frac{2(500)^2(1.1)^{-n}x}{[(1.1)^{-n}x + 500]^3} dx &= (1.1)^n \int_0^{600(1.1)^{-n}} \frac{2(500)^2y}{(y + 500)^3} dy \\ &= (1.1)^n \left[\int_0^{600(1.1)^{-n}} \frac{2(500)^2}{(y + 500)^2} dy - \int_0^{600(1.1)^{-n}} \frac{2(500)^3}{(y + 500)^3} dy \right] \\ &= (1.1)^n \left[\frac{2(500)^2}{(y + 500)} \Big|_0^{600(1.1)^{-n}} - \frac{(500)^3}{(y + 500)^2} \Big|_0^{600(1.1)^{-n}} \right] \\ &= (1.1)^n \left[\left\{ \frac{2(500)^2}{(500)} - \frac{2(500)^2}{(600(1.1)^{-n} + 500)} \right\} \right. \\ &\quad \left. - \left\{ \frac{(500)^3}{(500)^2} + \frac{(500)^3}{(600(1.1)^{-n} + 500)^2} \right\} \right] \\ &= (1.1)^n \left[1000 - \frac{1000}{(1 + 1.2(1.1)^{-n})} - 500 + \frac{500}{(1 + 1.2(1.1)^{-n})^2} \right] \\ &= \frac{1200}{(1 + 1.2(1.1)^{-n})} - \frac{600(2 + 1.2(1.1)^{-n})}{(1 + 1.2(1.1)^{-n})^2} = \frac{720(1.1)^{-n}}{(1 + 1.2(1.1)^{-n})^2}. \end{aligned}$$

By putting the two terms together, we have

$$E(Y_n) = \frac{720(1.1)^{-n}}{(1 + 1.2(1.1)^{-n})^2} + \frac{600}{[1 + 1.2(1.1)^{-n}]^2} = \frac{600}{1 + 1.2(1.1)^{-n}}.$$

- (iv) As n goes to infinity, the insurer's expected claim payment (obtained above) approaches 600.

This can also be argued from the fact that as claim amounts get inflated, virtually all claim amounts exceed the retention limit 600.

[This alternative explanation should fetch full credit.]

- (v) From part (ii), the expected claim size is seen to go to infinity as n goes to infinity. On the other hand, the insurer's expected claim payment saturates to 600, as found in part (iv). Therefore, the direct insurer's share goes to zero.

[10]

[TOTAL MARKS – 100]
